Filtered historical simulation: VaR 99% of 100 k€ for 21 Oct 2026 (1.0% of the book); 38 exceptions in 3,823 days for 38 expected

The method recommended by the backtest is selected; the slicer switches method. Each VaR is computed at the close for the next trading day.

Date

02/01/2008 | 21/10/2026

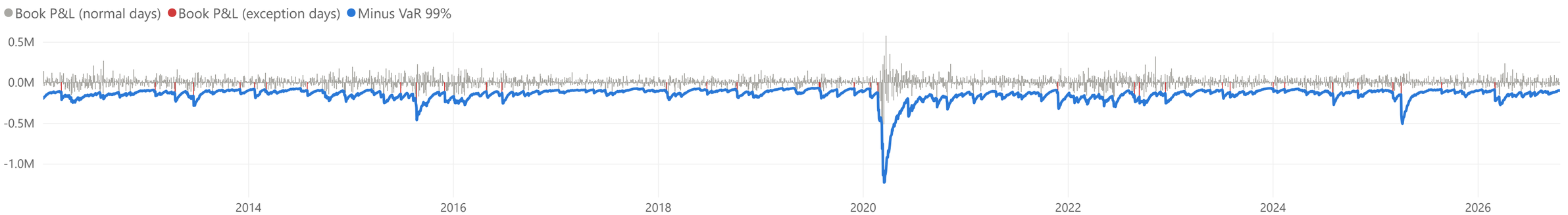
Method

Filtered historical simulation

VaR 99% (latest)	ES 97.5% (latest)	VaR 99% (latest), % of Book	Exceptions 250d	Traffic Light	Latest VaR Date
99,522	118,711	1.00%	1	Green	21 Oct 2026

Daily P&L of the book against minus the VaR of the previous close

Red days lost more than the VaR. One axis: the line is the VaR drawn as a loss.

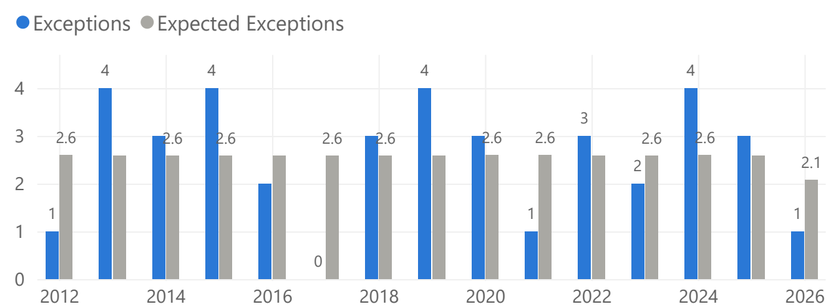


Backtest of every method, computed in Python

Tinted p-values are below 5%: the test rejects the method. Zones: Basel traffic light on 250 days.

Method	Backtest Period	Exceptions	Expected Exceptions	Kupiec p	Christoffersen p	Conditional Coverage p	Worst Zone

Exceptions by year against the 1% expected



GJR-GARCH(1,1) forecasts the book's volatility best (QLIKE 0.399 against 0.427 for HAR); XGBoost leads on the three equity indices

QLIKE (lower is better) and Diebold-Mariano tests against HAR, computed in Python on 2,269 out-of-sample days from 2018; GJR-GARCH feeds the VaR.

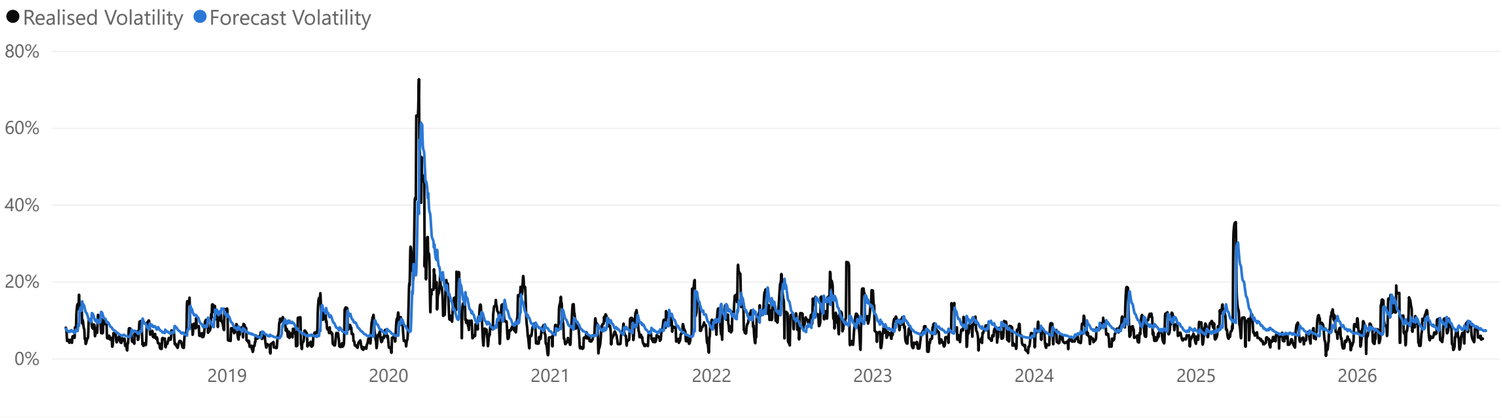
Series: Book

Model: GJR-GARCH(1,1)

Evaluation Period: 2018-2026

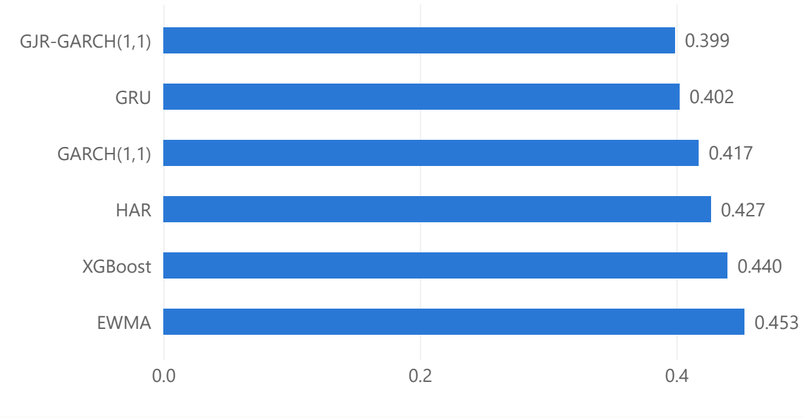
Forecast against realised volatility over the next 5 days, annualised

Series and model chosen above; forecasts made at each close from 2018.



QLIKE by model for the selected series

Lower is better.



QLIKE of every model on every series

Empty: HAR-range needs open, high, low and close, which the bond, the dollar and the book lack.

Series	EWMA	GARCH(1,1)	GJR-GARCH(1,1)	GRU	HAR	HAR (range)	XGBoost
Book	0.453	0.417	0.399	0.402	0.427		0.440
CAC 40	0.541	0.491	0.467	0.512	0.515	0.450	0.437
Euro Stoxx 50	0.534	0.494	0.464	0.503	0.518	0.466	0.452
S&P 500	0.499	0.441	0.413	0.436	0.456	0.444	0.404
Gold	0.441	0.419	0.414	0.551	0.427	0.420	0.472
10-year euro zero-coupon bond	0.339	0.332	0.332	0.391	0.344		0.391
USD cash	0.368	0.351	0.352	0.415	0.373		0.402

Diebold-Mariano against HAR for the selected series

Negative statistic: lower losses than HAR. Tinted: significant at 5%.

Model	QLIKE	DM Statistic vs HAR	DM p vs HAR
GJR-GARCH(1,1)	0.399	-1.75	0.080
GRU	0.402	-1.43	0.153
GARCH(1,1)	0.417	-1.12	0.262
HAR	0.427		
XGBoost	0.440	0.64	0.524
EWMA	0.453	1.69	0.090

S&P 500 ×5.3 and gold ×4.8 since January 2008, euro area equities ×1.4; equity volatility at 0.6 times its 2008-2026 average

Headline figures as of 21 Oct 2026 (latest values: S&P 500 and Gold 20 Oct). The snapshot, index and volatility follow the date and instrument filters.

Data as of

21 Oct 2026

Date

02/01/2008 — 21/10/2026

Instrument

All

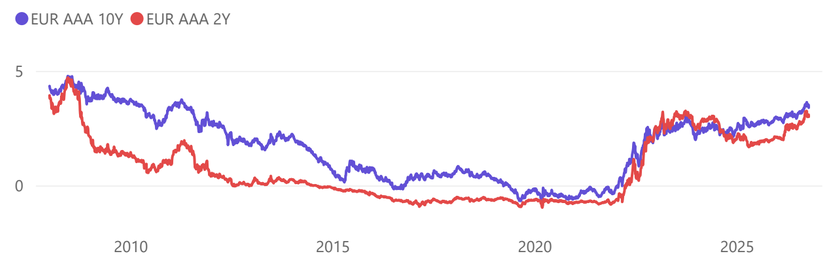
Snapshot over the selected dates

Volatility 22d vs Average below 1 means calmer than usual. Drawdowns count highs from January 2008.

Instrument	Return	Volatility (annualised)	Volatility 22d (latest)	Volatility 22d vs Average	Worst Day	Max Drawdown
S&P 500	+432.9%	19.8%	10.1%	0.62	-12.0%	-53.3%
Gold	+383.4%	18.4%	17.3%	1.02	-10.8%	-44.4%
Euro Stoxx 50	+43.2%	21.9%	12.2%	0.63	-12.4%	-58.2%
CAC 40	+42.7%	21.4%	11.8%	0.63	-12.3%	-54.6%
EUR/USD	-23.5%	8.9%	5.4%	0.66	-4.6%	-40.2%

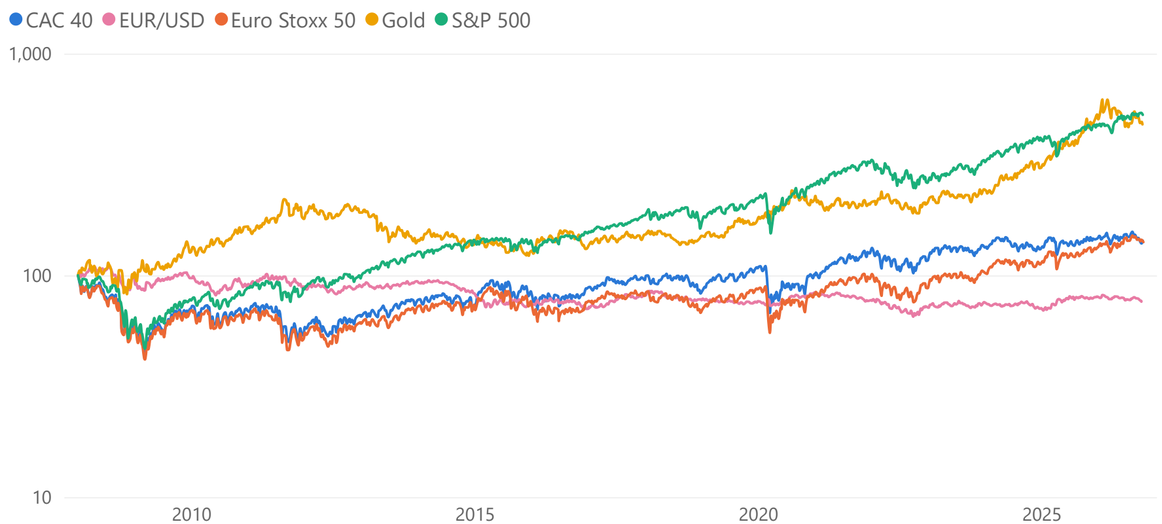
Rates, in percent

Negative at times, Jun 2016 to Mar 2022; low −0.82% (09 Mar 2020).

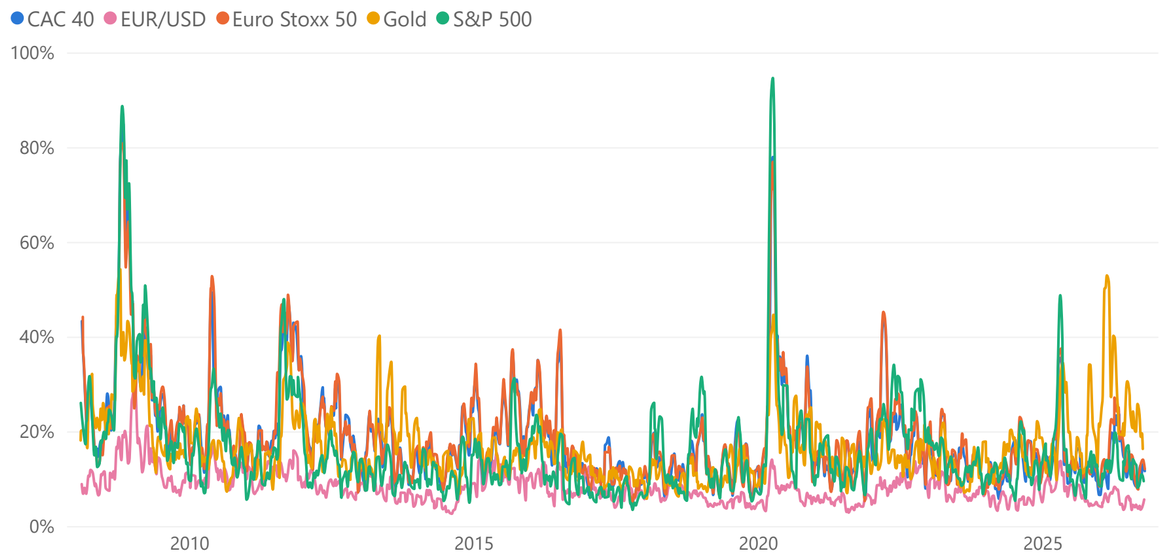


Index levels rebased to 100 on the first selected day

Logarithmic scale: equal distances are equal percentage moves.



Volatility of daily returns over the last 22 trading days, annualised



Euro bonds no longer offset equity losses: Euro Stoxx 50 returns and 10-year yield changes correlate at −0.47 over the last 250 days, against +0.20 since 2008

Correlations use daily returns, and daily changes for the yield, on the days both series moved.

Period

Full history

Instrument A

Euro Stoxx 50

Instrument B

EUR AAA 10Y

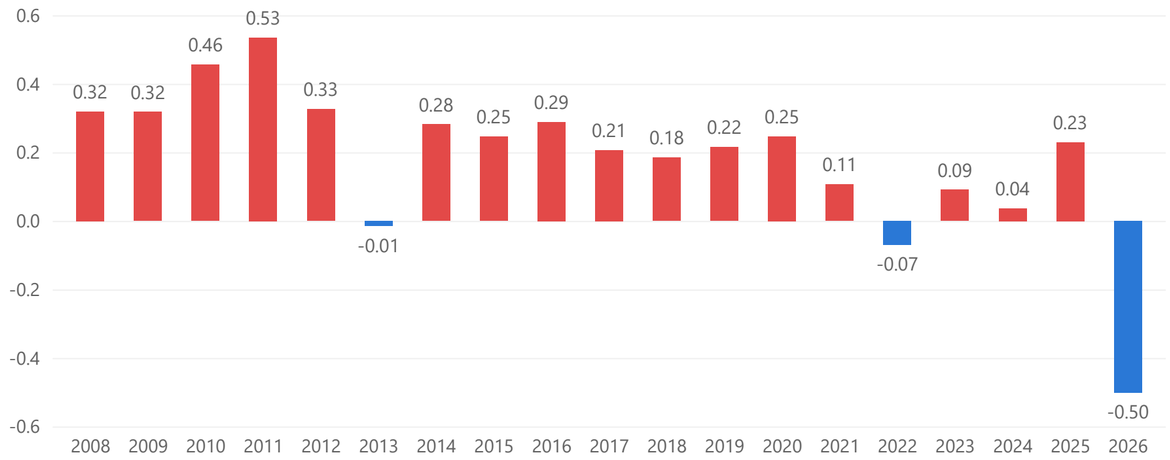
Correlation of daily moves over the selected period

Blue: the pair moves in opposite directions. Red: it moves together.

Instrument A	CAC 40	EUR AAA 10Y	EUR/USD	Euro Stoxx 50	Gold	S&P 500
CAC 40	1.00	0.21	0.17	0.97	0.02	0.58
EUR AAA 10Y	0.21	1.00	0.11	0.20	-0.12	0.09
EUR/USD	0.17	0.11	1.00	0.18	0.21	0.12
Euro Stoxx 50	0.97	0.20	0.18	1.00	0.03	0.59
Gold	0.02	-0.12	0.21	0.03	1.00	0.04
S&P 500	0.58	0.09	0.12	0.59	0.04	1.00

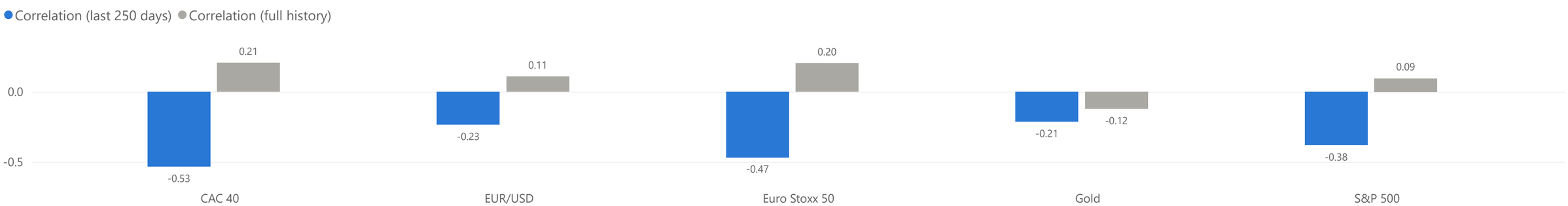
Correlation of the selected pair, by calendar year

2026 is the year to date.



Correlation with EUR AAA 10-year yield changes: last 250 days against the full history

Above zero, the bond gains when the asset falls and offsets its losses; below zero, both fall together.

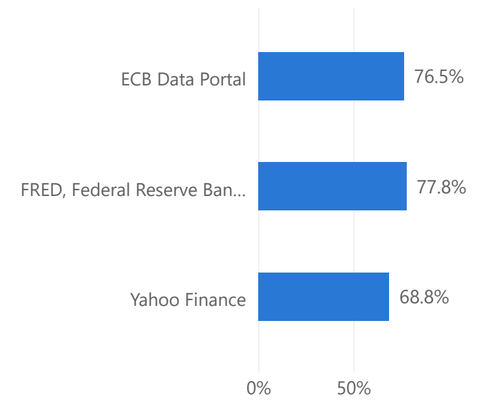


Latest quality run: no critical failure; 16 warnings, 12 of them missing trading days or late series

Run of 22 Oct 2026 09:36 UTC. Rules run after each download; a critical failure stops the pipeline, a warning is reported.

Critical Failures (latest run)	Warnings (latest run)	Series Late (latest run)	Last Quality Run (UTC)
0	16	6	22 Oct 2026 09:36

Share of checks passed, latest run



Flagged observations by rule and series, latest run

Empty: the rule does not apply to the series.

Rule	CAC 40	EUR AAA 10Y	EUR AAA 2Y	EUR/USD	EUR/USD (Yahoo)	Euro Stoxx 50	Gold	S&P 500	US 10Y-2Y	VIX
duplicate_key	0	0	0	0	0	0	0	0	0	0
fx_cross_source					16					
large_move	0			0	1	0	0	0		
late_data	0	1	1	0	1	0	1	1	0	1
missing_trading_day	6	10	10	0	29	98	3	0		0
non_positive_price	0			0	0	0	0	0		0
schema	0									
stale_value	0	0	0	0	1	0	0	0	88	0
yield_out_of_range			0	0						

Values flagged by the latest run

Yahoo values are left empty: the licence does not allow publishing them.

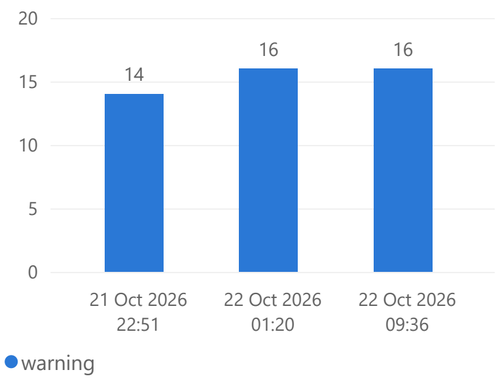
Observation Date	Instrument	Rule	Severity	Detail	Value
20 Oct 2026	EUR AAA 10Y	late_data	warning	last date 2026-10-20, expected 2026-10-21 or later	3.4401
20 Oct 2026	EUR AAA 2Y	late_data	warning	last date 2026-10-20, expected 2026-10-21 or later	3.0281
20 Oct 2026	EUR/USD (Yahoo)	late_data	warning	last date 2026-10-20, expected 2026-10-21 or later	
20 Oct 2026	Gold	late_data	warning	last date 2026-10-20, expected 2026-10-21 or later	
20 Oct 2026	S&P 500	late_data	warning	last date 2026-10-20, expected 2026-10-21 or later	
20 Oct 2026	VIX	late_data	warning	last date 2026-10-20, expected 2026-10-21 or later	
27 Aug 2026	US 10Y-2Y	stale_value	warning	same value 3 days in a row	
13 Aug 2026	US 10Y-2Y	stale_value	warning	same value 3 days in a row	
25 May 2026	Euro Stoxx 50	missing_trading_day	warning	no data on a XPAR session	

How late each series is, in trading sessions

No last date: the source delivered nothing.

Instrument	Last Date	Expected Date	Sessions Late
CAC 40	21 Oct 2026	21 Oct 2026	0
EUR AAA 10Y	20 Oct 2026	21 Oct 2026	1
EUR AAA 2Y	20 Oct 2026	21 Oct 2026	1
EUR/USD	21 Oct 2026	21 Oct 2026	0
EUR/USD (Yahoo)	20 Oct 2026	21 Oct 2026	1
Euro Stoxx 50	21 Oct 2026	21 Oct 2026	0
Gold	20 Oct 2026	21 Oct 2026	1
S&P 500	20 Oct 2026	21 Oct 2026	1

Failed rules per quality run



Market Risk Monitor: daily risk of a 10 M€ multi-asset book

Read me: what the report shows, where the data comes from and how the figures are defined.

Purpose
Daily market risk of a fictional 10 M€ multi-asset book (CAC 40, Euro Stoxx 50, S&P 500, gold, a 10-year euro zero-coupon bond and USD cash, constant notionals), built from free data by a scheduled Python pipeline.

Pages
Risk: today's VaR and ES of the book, its exceptions and the backtest of five VaR methods.
Models: volatility forecasts against realised volatility, QLIKE and Diebold-Mariano tests.
Market: returns, volatility, drawdowns and rates of the book's assets.
Correlation: how the assets move together, by period.
Data quality: results of the quality gate and freshness of every series.

Definitions
VaR 99%, 1 day: the loss exceeded on 1% of days, computed at each close for the next trading day.
Expected Shortfall 97.5%: the average loss on the worst 2.5% of days.
Exception: a day whose loss exceeds the VaR computed at the previous close.
Basel traffic light: exceptions over the last 250 days; green 0 to 4, yellow 5 to 9, red 10 or more.
QLIKE: loss function for variance forecasts, robust to noisy realised measures; lower is better.
Diebold-Mariano: test of equal forecast accuracy between two models.

Last market data

21 Oct 2026

VaR computed for

21 Oct 2026

Last quality run (UTC)

22 Oct 2026 09:36

Data and refresh
Yahoo Finance (equity indices, gold), ECB Data Portal (EUR/USD reference rate, AAA yields) and FRED (VIX, US 10y-2y). Raw Yahoo levels are not redistributed. The report reads the gold Parquet files through the DataFolder parameter (Transform data, Edit parameters), then Refresh.
Code, notebooks and methodology: github.com/MedChahb/market_risk_monitor